



F1 Bus A F2 Line mid F3 Bus B F4 Transformer HV F5 Bus C (external)

$$Z_{\text{src}} = j0.10 \text{ pu} \quad Z_{\text{line}} = j0.05 \text{ pu} \quad Z_{\text{tr}} = j0.08 \text{ pu} \quad Z_L = 2.0 \text{ pu} \quad (100 \text{ MVA, } 33 \text{ kV base})$$